- If 'a' and 'b' are any non-zero numbers and 'm' and 'n' are natural numbers, then

  (i) $a^m \times a^n = a^{m+n}$ (Product rule)

  (ii) $a^m \div a^n = a^{m-n}$, $m>n$ (Quotient rule)

  (iii) $\left(a^m\right)^n = a^{m\times n}$ (Power rule)

  (iv) $(a\times b)^m = a^m \times b^m$

  (v) $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

- For the base number whose unit digits are 0,1,5 and 6, the unit digit of a number corresponding to any positive exponent remains unchanged.
- For base ending with 4, the unit digit is 4 for odd power and is 6 for even power. Similarly, for base ending with 9, the unit digit is 9 for odd power and is 1 for even power.
- The largest power of a variable in an expression is called its degree. If it has more than one variable, then one has to take the sum of the powers of variables in each term and take the maximum of all these sums.

ICT Corner

Step-1 : Open the Browser type the URL Link given below (or) Scan the QR Code. GeoGebra work sheet named "Algebra" will open. There is a work sheet named "Law of Exponents"

Step-2 : Move the sliders a, m and n, observe the results and practice the laws.

Expected outcome



Step 1



Step 2



Browse in the link

Algebra: https://www.geogebra.org/m/f4w7csup#material/ab5ra9uf
or Scan the QR Code.

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